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**Research Article**

# Yttrium cuprates modification in linear generator application for power generation

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## Abstract

Yttrium barium copper oxide (YBCO) is used for special applications in linear generators because of its excitation loss, lower weight, and higher efficiency. These qualities enable the compound to operate better than the conventional copper wire coil in the stator unit of the linear generator. However, the continuous use of YBCO in linear generators has a fundamental challenge that affects industrial production and material stability after prolonged use. This paper seeks to sustain the adoption of YBCO by improving its quality for linear generator applications. The yttrium cuprates modification (YBYbCO) was synthesized using the solid-state reaction technique by doping YBCO with ytterbium. The crystalline structure, microstructural pattern, and stability of the new sample were adequately measured and found to be structurally stable to ensure durability. It was reported that applying YBYbCO in the linear generator would lead to a 200% increase in energy generation. The higher number of particles and lower individual particulate resistance enable it to withstand chemical pressure, thereby prolonging the lifespan of the linear generator.

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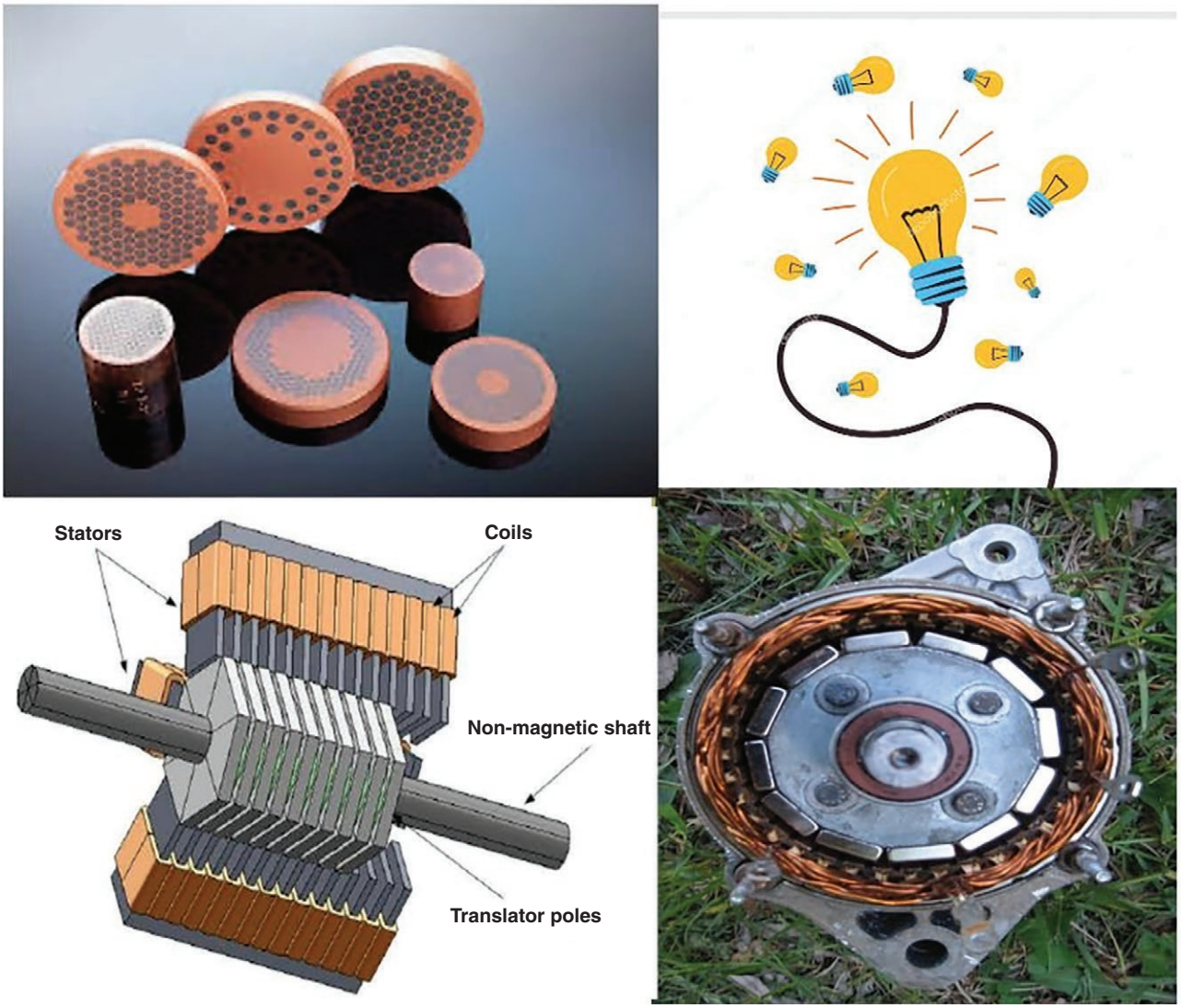
## Graphical Abstract




**Keywords:** microstructure; YBCO; structure; linear generator; cuprates; clean energy; SDG7

## 1. Introduction

The linear generator is a device used to convert mechanical energy into electrical energy. The source can be sea waves, fuel, wind, steam, tidal waves, and hydro-systems [1–3]. The main advantage of the linear generator is the absence of rotary parts such as a crankshaft and connecting rods [4]. In a standard engine, rotary parts contribute to friction losses that lower the efficiency. The most important part of a linear generator is the stator unit because all fluxes generated by permanent magnets are expected to flow, crossing the coils to convert the mechanical energy into electrical. Most modifications of linear generators are carried out in the stator units. For example, the old technology for linear generators makes use of round wire. Then, copper wire was replaced with superconducting wire to obtain a better coil fill factor [4]. The current-carrying capacity of copper wire, which is made up of 30 turns per coil, was configured at 27.5 A. In a bid to improve the efficiency of linear generators, the copper wire was substituted with a superconductor. Hossain *et al.* [5] reported that superconductor substitution in linear generators had improved the output power to about three times the output power of copper wire. There would be lower heat production as a copper winding is more compact for transferring heat by conduction.

Superconductors became the preferred choice for constructing linear generators [5, 6]. The superconductors are made into wires to replace the copper wire. Superconducting wires have a multi-filamentary design that allows each filament to be continuously arranged in a high-conductivity matrix—the sizes of the filament range from a few to some tens of micrometers in diameter. The high-conductivity matrix helps to transfer heat and current away from localized regions of the filaments. The disadvantage of the superconducting wire is the cost of cooling/refrigeration. Akune *et al.* [7] reported that the shortcomings in the filament, matrix and filament content factor significantly deformed the alternating current (AC) susceptibility curves. This result shows that there may be other underlying issues with the prolonged use of superconducting wire in linear generators. Most engineers have proposed yttrium barium copper oxide (YBCO) for linear generators, which is the main reason why it has flooded the market in

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recent times. When the cost of maintenance becomes high, users' interest may diminish. This research is geared to producing a test sample of modified YBCO to boost high efficiency and longevity. The modified YBCO in this context is YBCO that has been doped with ytterbium.

Doping YBCO with rare-earth elements (such as Nd, Sm, Eu, Gd, Dy, Ho, Er, Tm, Yb, and Lu) is not unknown in superconductivity. Different rare-earth elements have varying ionic radii and valence states, which can result in diverse effects on the YBCO crystal structure and electronic properties. It has been reported that doped YBCO improves upon the critical temperature ($T_c$) and current density ($J_c$) [8, 9] because the dopant that modifies the crystal structure and gives additional charge carriers can optimize the pairing interactions required for superconductivity. Also, rare-earth dopants sometimes create lattice distortions that impact the oxygen content within the $CuO_2$ planes [10, 11]. Hence, the choice of a rare-earth element and its concentration can be tuned to achieve the desired improvements in the superconducting properties. For example, [12] and [13] established that rare-earth ions can create pinning centers that enhance the flux-pinning capabilities of the YBCO material. Hence, improved flux pinning increases critical current densities and better performance in magnetic fields, which is crucial for practical applications.

The scope of the research is to focus on the test sample of the modified YBCO. The structure of the YBCO cuprate is somewhat dynamic and unpredictable, considering the previous literature [14–19]. The use of rare-earth metals to dope the YBCO structure has shown the following results among many. First, it enhances the thermodynamic stability of the sample [20]. It enhances magnetic flux pinning due to cation inter-substitution effects [21]. It improves the $T_c$ due to local strain fields formed by the rare earth due to the differences in their ionic sizes [17, 22]. In this research, the test sample was characterized to comprehend its crystalline structure, microstructural pattern, and stability under harsh conditions.

## 2. Uncertainties in YBCO

It is a fact that various doping proportions in YBCO are accompanied by a change in the crystal structure [23]. However, it is unclear what exactly influences the changes in YBCO when compared with most cuprates compounds [24]. Recent scanning tunneling microscopy and spectroscopy (STM/STS) experiments on YBCO have shown that damage to the atomic position or the electronic state initiates a dramatic temperature increase in the ultra-high vacuum [25]. Contrarily, since the structural anomalies of YBCO at room temperature are unknown, its corresponding changes at higher and lower temperatures may not be understood. This result means that, when the superconducting YBCO wire encounters abnormal conditions such as heating, the deformed AC susceptibility curves show that damage to the system is imminent.

This research paper proposes that the YBCO framework should be doped with ytterbium because of its inherent properties such as malleability, ductility, and relative stability. Its transformation temperatures are –13°C and 795°C [26]. The choice of the test sample (i.e. YBYbCO) is based on theoretical and experimental facts on ionic radii of rare earth (ytterbium), the proportion of constituents, preparation method, sintering temperature, and choice of sample characterization [18–23, 27–29]. The choice of ytterbium is to avoid the possibility of flattening the $CuO_2$ sheets or contraction of the in-plane Cu–O distance [30]. The selection of the test sample composition is based on three criteria. The first criterion is to understand the prospect of the Cu–Cu distance in a predefined lattice [27]. The second is to create layered cuprates with shorter coherence lengths because conventional YBCO does not allow supercurrent flow due to its peculiar grain boundary [31]. The third criterion is to assign a new role for BaO on the BaO/CuO chain layer. Based on the research objectives mentioned above, the test sample was prepared using a solid-state reaction, which is a synthesis method for obtaining polycrystalline material from solid reagents. It is preferred for the large-scale production of polycrystalline materials. It has been documented that the factors that influence solid-state reactions include the reactivity of the reagent, temperature, pressure, the environment of the reaction, surface area, and free energy change within the reacting reagents. The shortcoming of the solid-state reaction is the varying chemical reaction in a low surface area and incomplete reactions in some localized regions of the whole sample. In the modified sample, the surface area was made to be uniform by continuously grinding the compound at different stages of its preparation.

## 3. Linear generator application for high power generation

The linear generator has undergone extensive use in automobiles as an automaker. This development has heightened the clamoring to eradicate piston engines [3]. The linear generator in automobiles works by relative linear motion between the permanent magnets and the coils. This process extracts electric power from the combustion process. One of the limitations to its application to modern vehicles is the range and charging times. The advantages of the linear generator over the conventional engine include low noise emission, improved partial load efficiency, compact design, and balanced forces of inertia. Scientists have successfully deciphered the mathematical framework of the linear generator (Fig. 1).

The total flux flowing through the cross-sectional area of the modified YBCO can be calculated as:

$$\varphi = 2\pi \int B_z L_R dR \tag{1}$$

where $L_R$ is the thickness of the superconducting filaments and $B_z$ is the flux density. The induced voltage is derived from Faraday's law and can be estimated from the linkage flux in the superconducting as:

$$e = -N\frac{d\varphi}{dt} \tag{2}$$

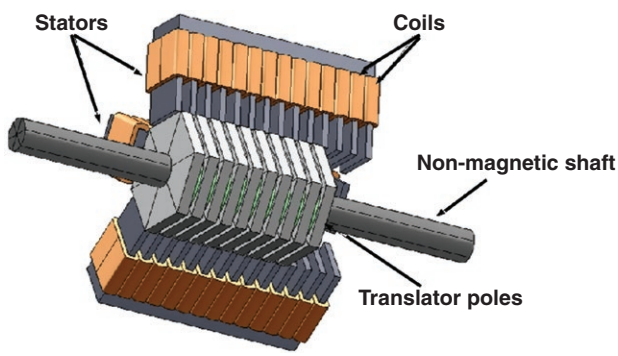


**Figure 1.** Typical linear generator design.

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where $N$ is the number of filaments and $d\varphi/dt$ is the rate at which the flux crosses the filaments. The cogging force maintains the alignment between the permanent magnet and the teeth of the stator. The core design of the linear generator is based on the science governing Equations (1) and (2). For example, Bashir *et al.* [2] proposed a linear generator in which the superconducting wires are made up of YBCO. The mathematical relationship of the power generated in the linear generator is given as:

$$P = \frac{V_t E_f}{X_d} sin\delta + \frac{V_t^2}{2}\left(\frac{1}{X_q} - \frac{1}{X_d}\right) sin2\delta \tag{3}$$

where $V_t$ is the terminal voltage, $E_f$ is the magnitude of the equivalent field voltage, $\delta$ is the power-factor angle, $X_d$ is the $d$-axis leakage reactance, and $X_q$ is the $q$-axis leakage reactance. It was proposed that the critical current, average thickness ($L_d$), and YBCO width should be 400 A, 0.1 mm, and 4 mm, respectively. The maximum tensile stress of YBCO is estimated to be >550 MPa. With such force on the filament, prolonged use would lead to thermally pressurized YBCO, which may likely lose its superconductivity.

The phasor diagram analysis of the linear generator is the primary requirement for calculating generator output power. However, the role of the superconductor is essential in optimizing the power output from the linear generator. The research on high-temperature superconducting linear generator (HTS-LG) systems has recently gained visibility [32, 33] and more insight have been acquired from the different experimental set-ups. The high-temperature superconductor (HTS) heat loss is an important factor that is considered in the HTS-LG system because it originates from the HTS flux flow. HTS flux flow in certain linear generator designs may generate so much heat that it may cause damage to the HTS wires. Aside from damaging the HTS wires, it hampers performance. Koshiba *et al.* [34] mathematically described the HTS heat loss as:

$$W = \sum_{z,r,\delta} \left\{ V_c I \left( \frac{I}{I_c(T, B, \theta)} \right)^{n(T,B,\theta)} r\Delta z \Delta r \Delta \delta \right\} \tag{4}$$

where $V_c$ is the critical voltage, $I$ is the HTS operating current, $I_c$ is the critical current, $n$ is the $n$-value of the HTS, $T$ is the operating temperature, $B$ is the applied magnetic flux density, and $\theta$ is the magnetic flux angle.

Most researchers have ignored the importance of the specification of the HTS material that would enhance higher performance. Until now, the level-set methods have been for numerical analyses of surfaces and shapes of the linear generator stator unit [35].

## 4. Preparation of over-doped yttrium cuprates using solid-state reaction techniques

The modified YBCO sample was prepared by using a solid-state reaction. The chemicals ($Y(NO_3)_3 0.6\ H_2O$, $(Ba(C_2H_3O_2)_2)$, $Yb(NO_3)_3.5H_2O$, $Cu(NO_3)_2.3H_2O$, citric acid, and ethylene glycol) were obtained in their purest form (>99.9%) from Sigma Aldrich.

The synthesis of the material is explained in detail in [36] and [37]. The stoichiometry (i.e. in moles) of the chemicals follows $Y_2BaYbCu_2O_{7-y}$, where 'y' may be in the range of 0.1–0.8. The calcination temperature of 850°C and a sintering temperature of 900°C were adopted. The calcination of the sample was performed three times in order to remove impurities. After each calcination, the samples were allowed to cool in open air. The resulting compound (i.e. black powder) was molded into pellets with thin thicknesses. The pellets were sintered and characterized accordingly.

## 5. Crystal and planar orientation of the modified/doped YBCO

The crystalline properties of the modified or doped YBCO were investigated using x-ray diffractometry (XRD). Fig. 2 presents the XRD analysis of the sample. It suggests that the sample had an orthorhombic structure with the most significant peak (2θ = 36.586°) showing a different optical plane, i.e. (103) plane, d[Å] =

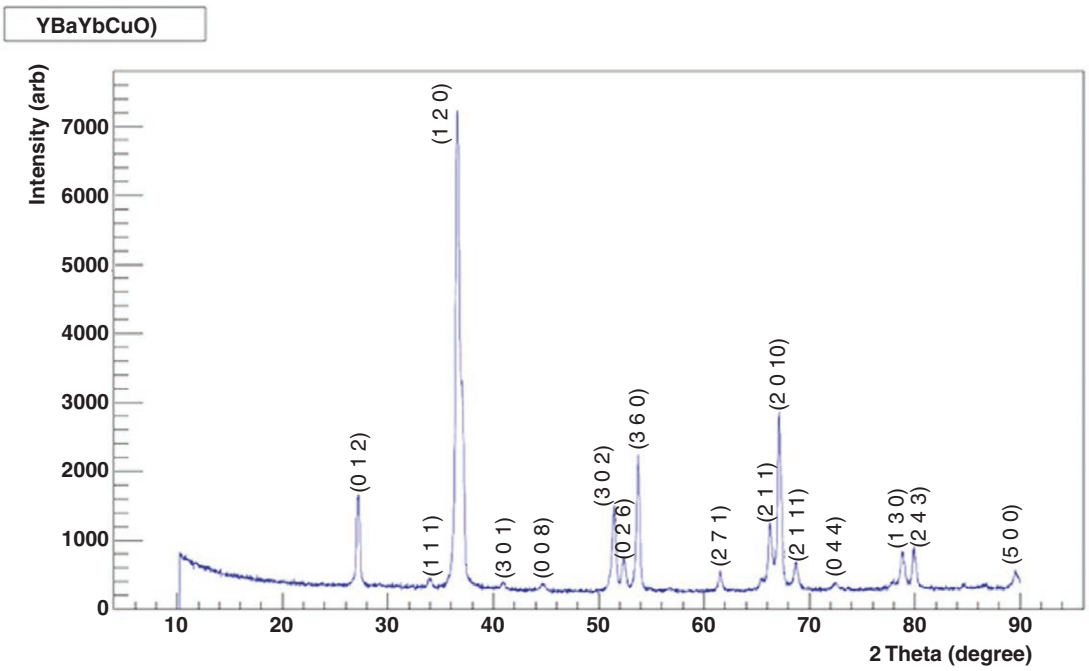


**Figure 2.** XRD patterns of YBYbCO.

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**Table 1.** Significance of Y and Ba via x-ray dispersion coefficients (λ = 0.154 059 8 nm) for individual elements.

| Element | f' | f'' | f_NT | μ/ρ (cm²/g) |
|---|---|---|---|---|
| Y | –3.014 35E–01 | 2.006 98E+00 | –9.385 10E–03 | 1.209 60E+02 |
| Ba | –1.242 15E+00 | 8.445 61E+00 | –1.252 70E–02 | 3.259 89E+02 |

**Table 2.** Estimate of non-destructive depth profiling.

| Ion energy | dE/dx elec. | dE/dx nuclear | Projected range | Longitudinal straggling | Lateral straggling |
|---|---|---|---|---|---|
| 10.00 keV | 9.632E–02 | 1.654E–02 | 500 A | 503 A | 388 A |
| 20.00 keV | 1.424E–01 | 1.309E–02 | 975 A | 787 A | 629 A |
| 30.00 keV | 1.803E–01 | 1.100E–02 | 1422 A | 989 A | 814 A |
| 40.00 keV | 2.128E–01 | 9.572E–03 | 1844 A | 1144 A | 965 A |
| 50.00 keV | 2.417E–01 | 8.524E–03 | 2244 A | 1269 A | 1094 A |
| 100.00 keV | 3.558E–01 | 5.707E–03 | 3998 A | 1660 A | 1543 A |
| 500.00 keV | 6.844E–01 | 1.876E–03 | 1.33 μm | 2459 A | 2785 A |
| 1.00 MeV | 6.903E–01 | 1.101E–03 | 2.31 μm | 2844 A | 3521 A |
| 1.50 MeV | 6.252E–01 | 7.982E–04 | 3.37 μm | 3172 A | 4165 A |
| 2.00 MeV | 5.616E–01 | 6.328E–04 | 4.56 μm | 3536 A | 4825 A |

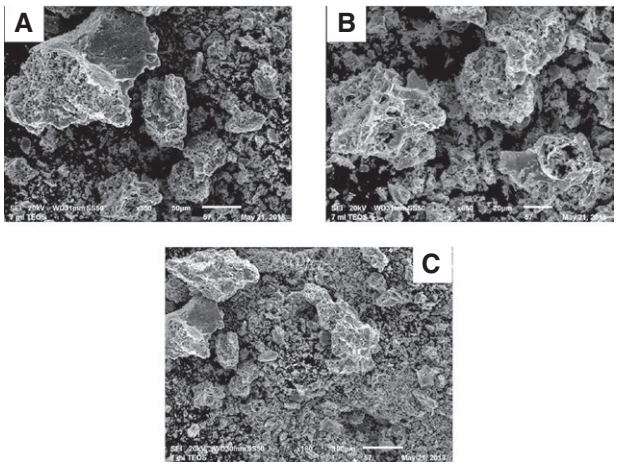


**Figure 3.** SEM images of YBYbCO.

2.4548, I/I0 = 969.35 and full width at half maximum (FWHM) = 0.1600 for all the samples. The range of the scan was between 9.096° and 91.006°. The Bragg–Brentano geometry (which describes the specimen displacement correction) is given as $T$ = (–s/R) = –0.00029. Characterization of the peaks was carried out, as the identified peaks were 97.16% [36, 38–40]. The peak residual of 2.84% may have been associated with impurities, as reported in [15]. The possible layer of the YBYbCO sample ranges from YO/$CuO_2$/YbO/$CuO_2$/YO/BaO layers and there is a need to understand the XRD coefficient of the elements, as shown in Table 1. The XRD coefficients Cu, Yb, and O are listed in [37]. This idea is salient for discerning elemental fluctuations in each layer. For example, the $\mu/\rho$ factor, which is referred to as the mass attenuation coefficient, corroborates the influence of earth-based superconducting phases in the unidentified peaks. The $\mu/\rho$ factor is the attenuation per unit mass that characterizes how easily a mass of material can be penetrated by a beam of light, sound, x-ray, particles, or other energy or matter. Contrary to [41, 42], the possibility of a non-superconducting impurity phase associated with the BaO surface may not be inferred because $\mu/\rho$ in Table 2 suggests that, in the case of YBYbCO, the surface cleavage may not be BaO or CuO chain layers only. These results may have emerged because the $\mu/\rho$ factor for barium and copper in the YBYbCO compound are the highest and lowest, respectively. Secondly, the $\mu/\rho$ factors for yttrium and ytterbium are somewhat close; hence, phase fluctuation between the elements may be significant in the YBYbCO lattices. Thirdly, the elemental roles of yttrium and ytterbium in the bi-atomic superconductors, i.e. $YB_6$ [43] and $C_6Yb$, are noteworthy, especially when both elements are involved in a reaction. From the existing database, the unit cell of the sample has an average $a$ = 7.1320 Å, $b$ = 12.1810 Å, $c$ = 5.880 Å. However, the computational result of the unit cell of the YBYbCO compound is given as $a$ = 7.13200 Å, $b$ = 12.18100 Å, $c$ = 5.65800 Å.

## 6. Microstructural properties of the modified/doped YBCO

Scanning electron microscopy (SEM) was used to investigate the surface microstructural signature of the YBYbCO compound. The granular morphology of the SEM images is presented in Fig. 3. The microstructural pattern of the modified YBCO shows that the sample has considerable grain boundaries that allow heat losses when used in a linear generator. The atomic position of the modified YBCO 3D is presented in Fig. 4. The formation of an Abrikosov lattice (hexagonal pattern) at the center of the compound was observed. The Abrikosov lattice is peculiar to the Y-123 (yttrium barium cuprates) formation. Hence, electrons are trapped in cage-like boundaries. The features of this boundary show that the lattice of the modified YBCO is not a perfect square lattice [14] nor a triangular lattice system [41].

It was observed that the strain of YBYbCO does not move along the '$a$' and '$b$' axes, as reported in [42]; instead, they tend to translate towards the '$a$' and '$c$' axes. Therefore, it is evident that the compound would experience lattice mismatch due to chemical pressure [44, 45]. This property is excellent for accommodating the stress on the superconducting wires. More so, experiments have proven that sub-lattice mismatch has a significant influence on the transition temperature [45, 46]. The corresponding ionic pattern within the atomic position of YBYbCO that causes

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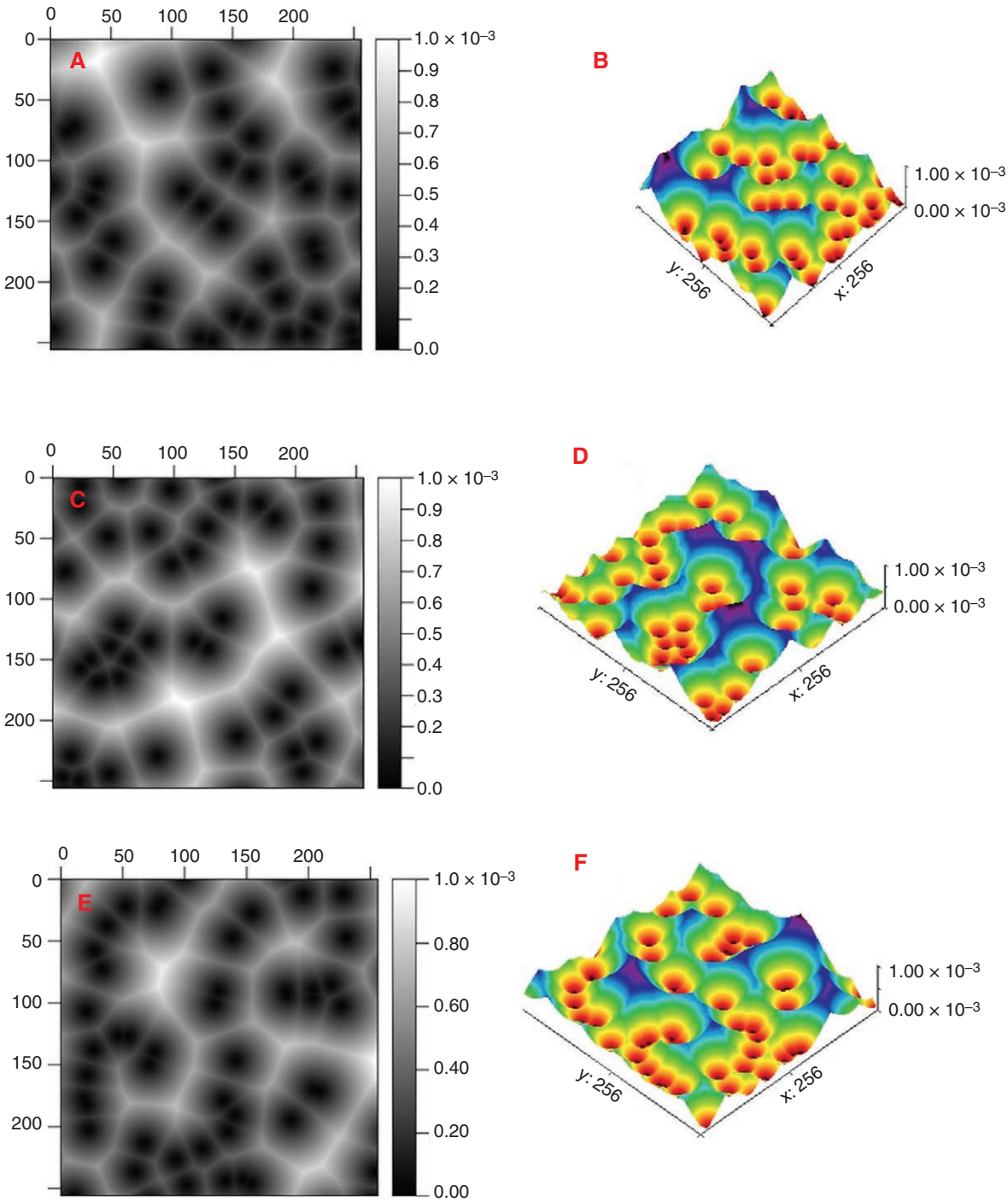


**Figure 4.** Atomic position in YBYbCO.

it to differ from Y-123 compounds was further examined. The copper oxygen has an 8-fold coordination with the ionic bond $b_1$, indicating that the Y cation resides between its double layers. The double $CuO_2$ layers oriented in the (111) plane are evidence of the ionic bond $b_2$. This scenario is likely to occur when it lies within the *a*–*b* axes. Hence, along the *a*–*b* axes, the misorientation of YBYbCO is more dynamic at the ionic bond. For example, on the z-axis, the Y cation resides between the 4c site (double layers) of the copper oxide plane, which has an affinity for only O(3). Oxygen O(1) and O(2) lie in the 8d site and are not fully occupied as earlier postulated. It can be inferred that the YbO plane resides in the Wyckoff positions, i.e. the 8d site, where eight oxygen atoms in the ac plane are located [47].

The SEM images were cropped into four parts, labeled as sections 1–4. The surface was estimated for each SEM image (Fig. 4a–c) and is presented in Fig. 5. The surface microstructure for Fig. 3a–c is presented in Fig. 5a–c, respectively, while the corresponding size (*p*) is presented in Fig. 5b, d, and e. The focus of this section is to see how the modifications in the experimental set-up of solid-state reactions have affected the localization of the individual reagent. It is observed (from Fig. 5a and b) that Fig. 3a is the most incoherent. This result is because the prominent element,

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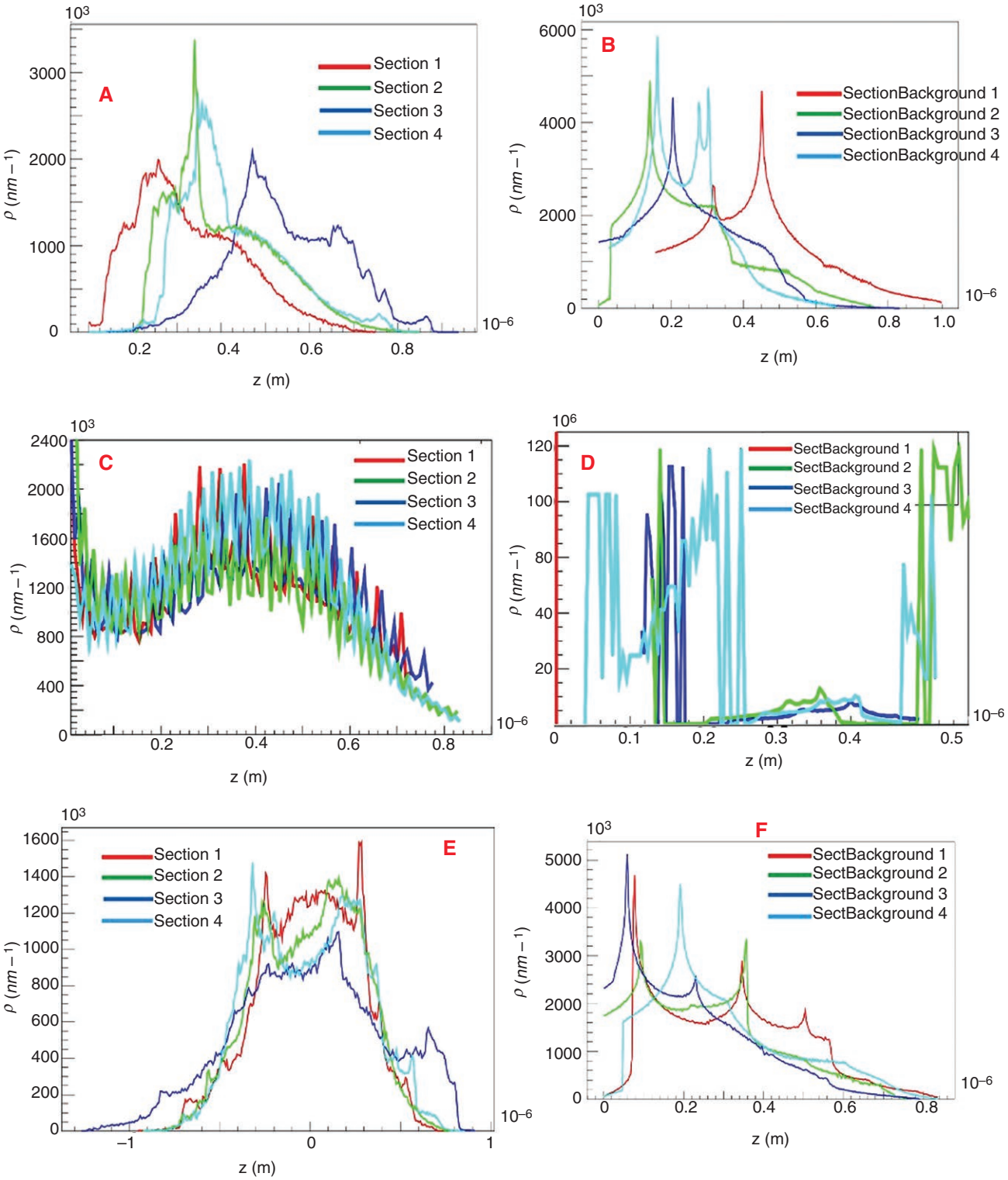


**Figure 5.** Interatomic length and size within the granular nature of YBYbCO.

i.e. barium in the form of $BaCuO_2$ has a wide x-ray dispersive coefficient that distinguishes it from ytterbium and yttrium. Aside from those mentioned above, the heterogeneous nature of Fig. 5a and b might occur via the random gap disorder associated with the dopant atoms [47].

On a broader view of Fig. 5, the copper oxide is proposed to be in the background of each image while the full compound analysis is presented in bulk. For example, Fig. 5a, c, and e show full analyses of each sample while Fig. 5b, d, and f show the respective backgrounds.

Like any cuprate superconductor, the atomic-scale mechanism in the $CuO_2$ plane influences the YBYbCO framework. Fig. 5c–e reveals that Fig. 2b and c are mainly ytterbium and yttrium. Fig. 5c–e shows an almost homogenous material that has close

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surface microstructures and sizes. The broad pattern (Fig. 5f) is evidence that there is a decrease in the distance between the apical oxygen and the $CuO_2$ plane. This view is contrary to the observations in [48].

## 7. Material stability testing using the ion bombardment technique

The stability of the sample was tested using the principle of the elastic scattering of energetic ions. Rutherford backscattering spectrometry (RBS) was adopted to examine the ion bombardment analysis of the sample. The projectile ion helium was projected at an energy of 2 MeV. The sample was fixed at 60° on a goniometer. Another advantage of using RBS was to examine the cation concentrations for Y, Ba, Yb, and Cu in the surface and near-surface regions [49, 50]. The non-destructive depth profiling of the sample is presented in Table 2. The non-destructive depth profiling is a Monte Carlo simulation method for the stopping and range of ions in matter. It was performed by using the binary collision approximation for 99 999 ions. Table 2 shows that the longitudinal and lateral straggling of ionic collisions in YBYbCO varies considerably. Hence, it is evident that the material is stable under harsh conditions.

Furthermore, it is observed that the electronic and nuclear properties of the YBYbCO sample increase and decrease, respectively, as the ion energy increases from 10 keV to 1.0 MeV. Also, fluctuating values of the electronic and nuclear properties are observed at between 1.0 and 2.0 MeV. This observation may have resulted from overlap caused by the high yields of heavier elements [51]. The projected range, longitudinal and lateral straggling, increases as the ion beam increases. The radial straggle fluctuates due to the nuclear output from the elastic scattering of energetic ions from each atomic nucleus. The total ionization in YBYbCO is 1958.8 keV/Ion, while the target and phonon damage are given as 1.86 and 39.4 keV/Ion, respectively. It may be inferred that the electron-to-phonon ratio in YBYbCO may be ~1:21. Mingo and Broido [52] reported that electron–phonon interactions have influenced the thermal conductivity of materials, as cases of strong interactions resulted in a reduced thermal conductivity because of the phonon scatter off electrons. Electron–phonon coupling can significantly influence the mechanical properties of the material. Galli and Natoli [53] also postulated that electron–phonon interactions have a significant influence on the ability of a material to absorb and dissipate energy, hence increasing the material toughness and ability to withstand mechanical stress. In this regard, some scientists have reported that materials with optimal electron–phonon coupling tend to exhibit enhanced hardness and elastic moduli [54, 55]. This is based on the above premise that we infer that YBYbCO is structurally stable enough to withstand the operational stress of the stator unit in the linear generator.

The RBS spectrum of the YBYbCO sample is presented in Fig. 6. The diagram also reveals the Z-element classification. The possibility of elemental phase interaction was explained by using the x-ray dispersive coefficient and was affirmed by the closeness of the heavy Z element, as presented in Fig. 6. The RBS spectrum was analysed by using the curve-fitting technique that is hinged on parameters such as Cal A, Cal B, Beam energy, detector FWHM, beam-detangle, in-plane tilt angle, out-of-plane tilt angle, straggling, pile-up, and constant background. It was observed that, when no parameter was used for the curve fitting, the YBYbCO composition was $Y_{6.7}Ba_{0.41}Yb_{3.37}Cu_{3.28}O_{15}$ at a Q-factor of 0.043, chi-square of 0.5489, and dMax of 78. Also, when all of the parameters listed in Table 1 were used to plot the fit, the elemental composition of YBYbCO was given as $Y_{6.8}Ba_{1.82}Yb_{6.88}Cu_{3.25}O_{15}$ at a Q-factor of 0.043, chi-square of 0.481, and dMax of 81.

The RBS compositional analysis has further affirmed that YBYbCO is structurally stable to withstand the stress in the stator unit of the linear generator. The success of YBYbCO has been proven on the laboratory scale. Hence, it is recommended that industrial-scale testing should be carried out for further study.

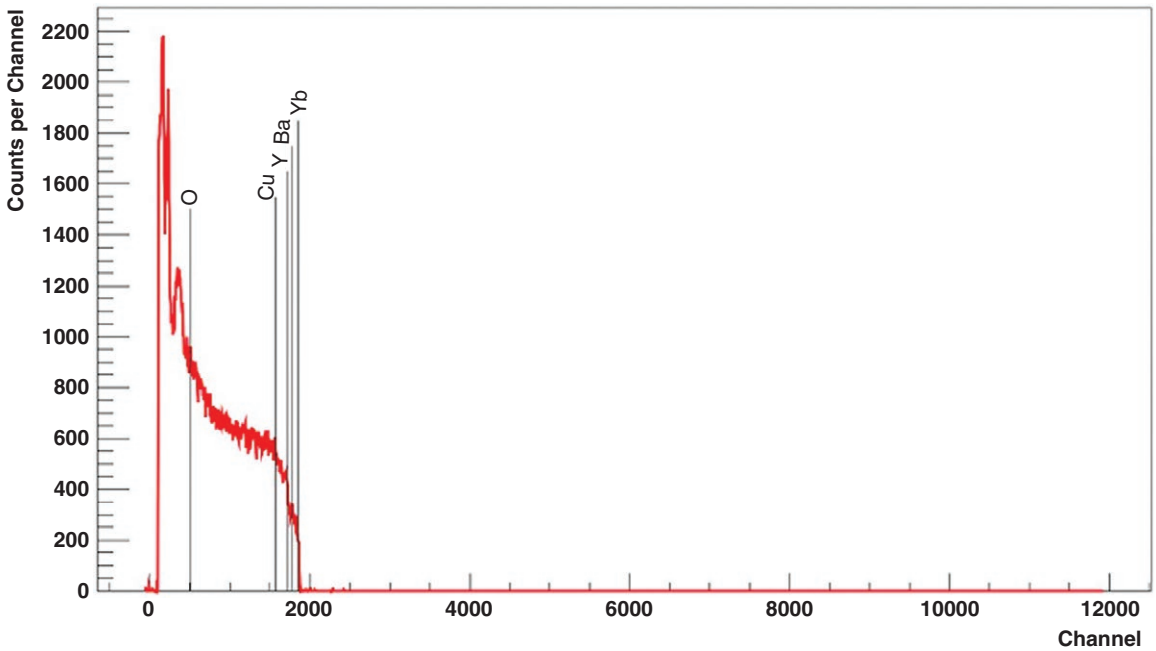


**Figure 6.** RBS spectrum of Z element.

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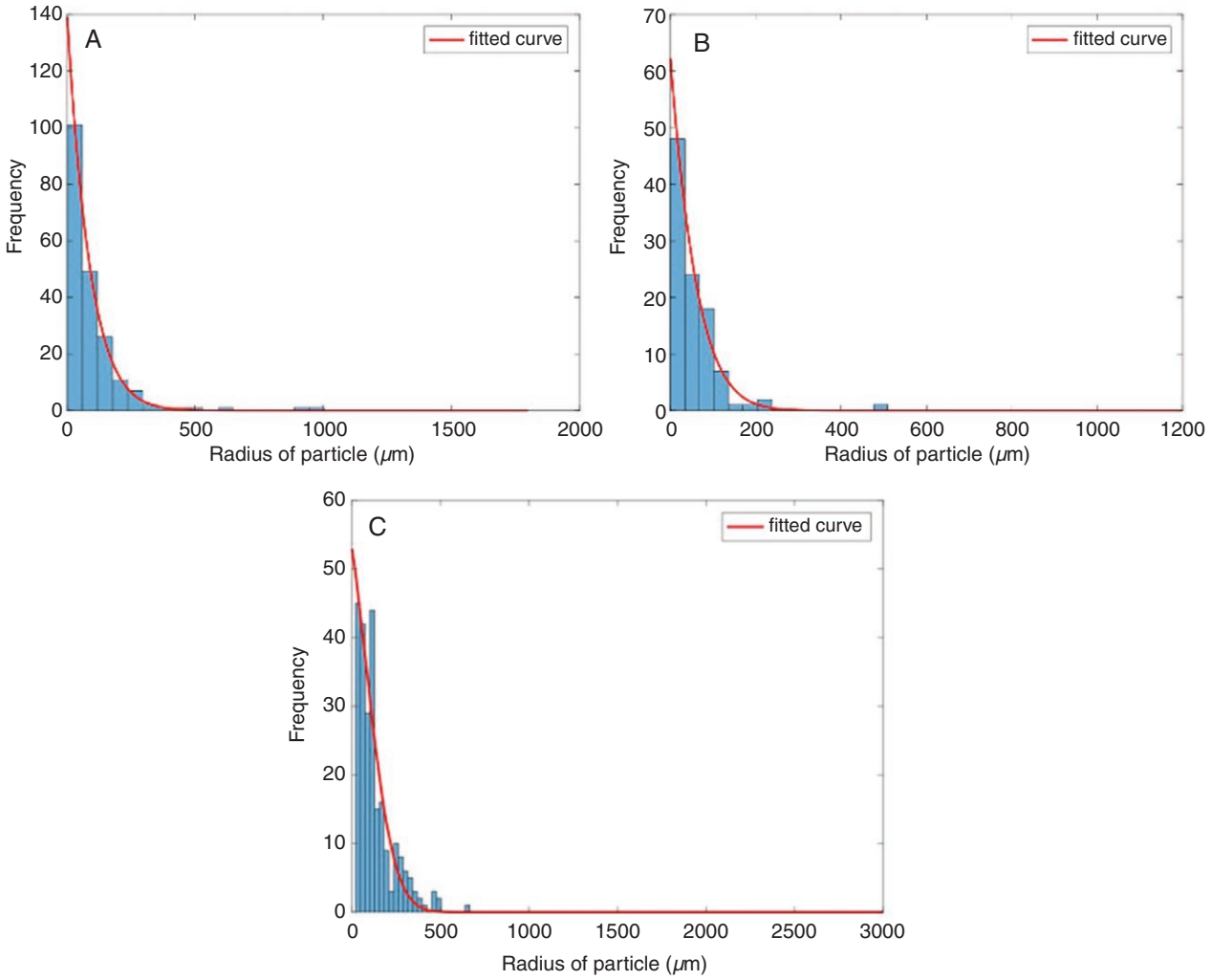


**Figure 7.** Radii of particulates of YBCO.

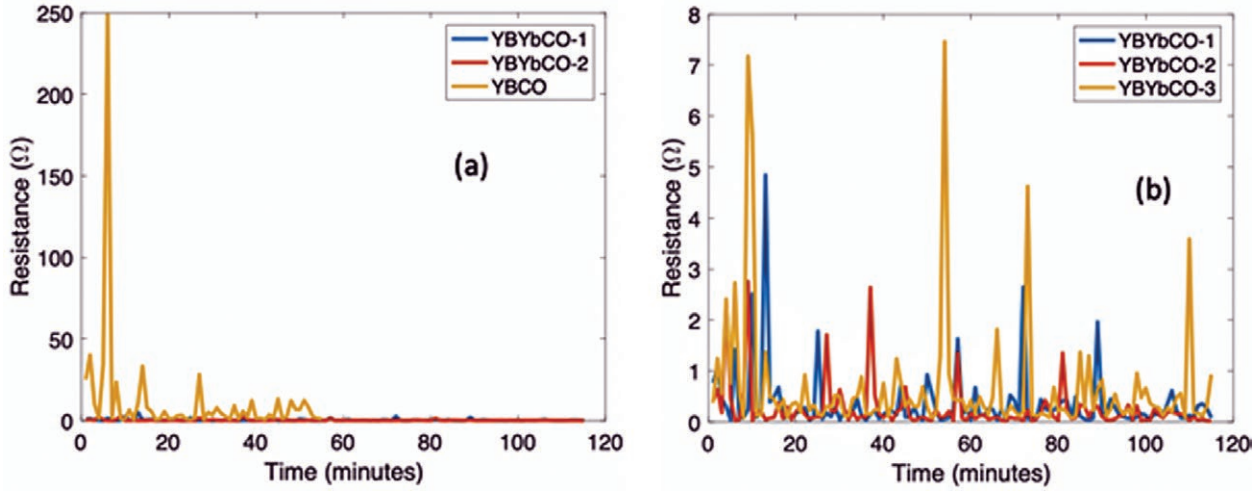


**Figure 8.** Fluctuating resistances due to prolonged force on the filament.

## Theoretical application of modification to the linear generator

Many researchers have experimentally worked on the determination of the electrical resistance of a powdered material [56–59]. The work of Hong *et al.* [57] was preferred in this work based on technical issues that include the unification of the measuring parameters. For example, the use of compressive force, base disc, etc. is dependent on the equipment and accuracies at the point of measurement. The electrical resistance was given as:

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$$R = r\left(\frac{4\pi}{3\phi}\right)^{\frac{1}{3}} - 2 \tag{5}$$

where $r$ is the particle radius and $\phi$ is the volume fraction of the filler in the composite. Though Equation (5) may apply for bulk samples, in this context, in which the radius of the particulate is significantly small, the modification to the equation becomes:

$$R = r\left(\frac{4\pi}{3\phi}\right)^{\frac{1}{3}} \tag{6}$$

The radius of the particulate was obtained from the SEM images (Fig. 3), as presented in Fig. 7. The average radii for the three images are given as 126.1, 92.1, and 208.2 μm. Contrary to YBYbCO, the average radii of the particulate of YBCO as calculated from [60] is 4.16 μm.

The resistance term in Equation (3) is given as:

$$R = \left(\frac{1}{X_d} sin\delta + \frac{1}{2}\left(\frac{1}{X_q} - \frac{1}{X_d}\right) sin2\delta\right)^{-1} \tag{7}$$

At critical temperature, it is assumed in this research that $\delta = 90°$ such that $R = X_d$. In this case, it is believed that:

$$R = X_d = r\left(\frac{4\pi}{3\phi}\right)^{\frac{1}{3}} \tag{8}$$

which will translate into:

$$P = \frac{V_t E_f}{r}\left(\frac{4\pi}{3\phi}\right)^{-\frac{1}{3}} \tag{9}$$

It was proposed that the critical current, average thickness ($L_d$), and YBYbCO width are given as 400 A, 0.1 mm, and 4 mm, respectively. The maximum tensile stress of YBYbCO is estimated to be >550 MPa. Based on the above estimation, we adapt the typical resistances of a linear generator that ranges from 12 to 30 Ω [61, 62]. Based on the calculation, i.e. by using Equation (8) at a volume fraction of 0.2, the expected resistance will be given as 0.269, 0.105, and 0.574 Ω. Contrary to the resistance of YBYbCO, the expected resistance of YBCO was calculated to be 11.48 Ω.

This result means that power generation using the modified YBYbCO would lead to an increase of >200% over the conventional routes. The only disadvantage of using a superconductor for the linear generator is the prolonged force on the filament, which creates thermal pressure on each superconductor bar in the linear generator. This event ultimately leads to the loss of superconductivity. The advantage of YBYbCO over YBCO is its stability during thermal pressure and how the resistance of the individual particulate behaves before it eventually breaks down. In this case, Fig. 8 demonstrates the resistance fluctuations of the superconductor bar in the linear generator.

Fig. 8a shows that YBCO would have distinctive resistance fluctuations compared with YBYbCO under thermal pressure from a linear generator. Secondly, YBCO has fewer particulates compared with YBYbCO, hence fewer particulates undergo thermal pressure. More significantly, the magnitude of the resistance fluctuation is higher in YBCO, which technically engenders system failure faster.

## Conclusion

The chemical and structural stability of the modified YBCO has been explained to be a perfect replacement for superconducting wires for linear generator applications. Hence, computational assumptions in estimating the surface and size of the HTS in a linear generator should be discouraged, as there are more technicalities in material variations. The strain (in this modified YBCO framework) does not move along the '*a*' and '*b*' axes, as reported by existing research. However, there was evidence of lattice mismatch at room temperature, as presented in the body of the work. The oxygen element (O (1) and O (2)) in the YBYbCO lies in the Wyckoff positions 8d site and is not fully occupied, as previously reported. This result causes the interatomic bond angles in the Y site to decrease along the *a*-axis and fluctuate along the *b*-axis.

The RBS technique adequately affirmed the stability of YBYbCO by the use of phonon analysis and compositional variations. The electron–phonon ratio in YBYbCO was ~ 50:1. There are local lattice fluctuation and atomic-scale mechanisms of the $CuO_2$ plane that control the charge reservoir in YBYbCO. This discovery is expected to guide experimental procedures on surface termination techniques to activate or optimize the topmost $CuO_2$ plane. The localized lattice structure within the bulk cuprates has been shown to have a significant influence on the entropy and electron transport in the YBYbCO compound. The success of YBYbCO has been proven on a laboratory scale. Hence, it is recommended that industrial-scale testing should be carried out for further study.

It was reported that applying YBYbCO in a linear generator would lead to a 200% increase in energy generation compared with conventional routes. Due to the higher number of particles and lower individual particulate resistance in YBYbCO, it could withstand chemical pressure, thereby prolonging the lifespan of the linear generator.

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## Author contributions

Moses Emetere (Conceptualization [Equal], Data curation [Equal], Formal analysis [Equal], Investigation [Equal], Methodology [Equal], Resources [Equal], Supervision [Equal], Writing—original draft [Equal], Writing—review & editing [Equal])

## Conflict of interest statement

The author declares that they have no conflict of interest.

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## Data availability

The data will be provided upon request.

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